Genotype-to-Rescue

A mechanism-first approved-drug repurposing strategy for BUB1B-associated Mosaic Variegated Aneuploidy

Proposal

Track 1 recovered the hackathon-confirmed BUB1B variant pair at rank 1. Track 2 translates that molecular signal into a falsifiable repurposing program. A genetic rescue benchmark establishes how much of the cellular phenotype is BUB1B-rescuable; approved drugs are then tested against two separable liabilities: ongoing chromosome instability (Arm A) and the cellular burden of established aneuploidy (Arm B).

BUB1B genotype → WT-BUB1B rescue benchmark → Approved-drug screen → Target engagement → Functional rescue

ARM A - REBALANCE	ARM B - CONTROLLED BUFFERING
Lead: fingolimod (FTY720). Tests whether SET/PP2A modulation can improve the kinetochore phosphatase balance linked to BUBR1 and reduce chromosome missegregation.	Leads: 4-PBA and arimoclomol. Tests whether a demonstrated proteostasis phenotype in aneuploid patient cells can be improved without increasing genomic instability.

Case-specific molecular foundation

The working model contains NM_001211.6:c.2210T>G (p.Leu737Ter), an established pathogenic loss-of-function allele, and c.3006T>G (p.Asn1002Lys), a rare candidate second missense allele that remains a VUS. Phase is unresolved. The project therefore treats BUB1B-associated MVA1 as a compelling molecular hypothesis while requiring genetic and functional resolution of the second allele.

Therapeutic thesis

Restoring WT BUBR1 is the mechanistic ceiling, not the Track 2 drug nomination. The repurposing question is whether an approved pharmacologic intervention can reproduce a meaningful component of that rescue - either by reducing new chromosome-segregation errors or by safely buffering a measured consequence of existing aneuploidy.

1. From genotype to a measurable rescue benchmark

BUBR1 is a spindle-assembly-checkpoint and kinetochore regulator. BUBR1-dependent recruitment of PP2A-B56 supports chromosome congression and kinetochore-microtubule regulation. BUB1B-mutated MVA patient cells show reduced BUBR1 and chromosome-congression defects, providing a direct cellular substrate for rescue experiments.

L737* / N1002K → BUBR1 abundance or function ↓ → kinetochore/checkpoint defect → missegregation ↑ → mosaic aneuploidy

Genetic rescue benchmark

WT-BUB1B re-expression is used as the positive mechanistic benchmark. Prior BUB1B-MVA work demonstrated that restoration of WT BUBR1 can restore checkpoint function, and that an intact BUBR1-B56 interaction is required for rescue of chromosome congression in BUB1B-mutated MVA cells. Independent mouse studies further show that sustained BubR1 overexpression can improve checkpoint and kinetochore-microtubule attachment defects while reducing aneuploidy and tumorigenesis. These findings support genetic restoration as a benchmark for biological reversibility, not as a proposed treatment in this Track.

Question	Assay	Interpretation
Is L737* expressed?	Allele-specific RNA; BUBR1 abundance	Distinguishes likely NMD from residual truncated-protein models.
Does N1002K alter BUBR1?	Matched-expression WT, N1002K, L737*, empty vector; abundance, turnover, localization	Defines stability versus functional defect and can downgrade N1002K if essentially normal.
Is the patient phenotype BUB1B-rescuable?	WT-BUB1B re-expression; checkpoint, B56 recruitment, alignment, missegregation	Defines the genetic rescue ceiling against which pharmacologic rescue is interpreted.

Key experimental principle

A drug is not credited for changing a surrogate alone. Pharmacologic rescue is interpreted relative to baseline and WT-BUB1B rescue, with chromosome missegregation per observed mitosis as the primary Arm A functional endpoint.

2. Systematic approved-drug prioritization

Approved and clinically used compounds were mapped to BUBR1 stability, PP2A/kinetochore regulation, proteostasis, redox/ISR biology, mTOR/autophagy and mitochondrial homeostasis. Prioritization considered mechanistic proximity, human-cell evidence, chromosome-instability or aneuploidy relevance, measurable target engagement, regulatory status, and genomic-safety liabilities. The screen ranks experiments, not predicted clinical efficacy.

Approved-drug space → mechanism mapping → human/CIN/aneuploidy evidence → safety filter → focused shortlist

Candidate	Role	Rationale	Disposition
Fingolimod	Arm A - REBALANCE	Reported SET/PP2A modulation provides an approved-drug bridge to the BUBR1-PP2A-B56 hypothesis.	Lead Arm A
4-PBA	Arm B - proteostasis	Human constitutional-trisomy fibroblast evidence links aneuploidy to proteostasis disturbance and shows reduction of aggregate burden with 4-PBA.	Lead Arm B
Arimoclomol	Arm B - heat-shock response	Approved chaperone-response modulator; mechanistically distinct proteostasis strategy.	Arm B comparator
DRF / DMF	Arm B - redox/ISR	Nrf2-linked stress-response pharmacology.	Phenotype-gated
Elamipretide	Arm B - mitochondria	Mitochondrial/cardiolipin-directed pharmacology.	Phenotype-gated
Aurora B modulation	Mechanistic control	Directly interrogates kinase-phosphatase balance affecting chromosome congression.	Control, not chronic lead

The focused program therefore tests one upstream-comparative benchmark (WT-BUB1B), one approved Arm A lead (fingolimod), and a mechanistically diverse Arm B mini-screen. Secondary candidates enter only when the corresponding patient-cell phenotype is present.

3. Arm A - pharmacologic rescue of chromosome-segregation fidelity

Fingolimod is prioritized to test whether pharmacologic relief of SET-mediated PP2A inhibition can produce functionally relevant phosphatase rebalancing when BUBR1-dependent kinetochore regulation is impaired. The critical question is not whether global PP2A changes, but whether target engagement reaches the mitotic machinery and reduces segregation error.

FTY720 → SET / PP2A engagement → kinetochore B56 readout → congression improves → missegregation ↓

Domain	Measurement	Advance criterion
Target engagement	PP2A activity; kinetochore PP2A-B56 signal	Biochemical engagement accompanied by kinetochore-relevant change.
Mitotic function	Alignment, lagging chromosomes, mitotic progression	Improvement without artificial arrest or suppression of mitotic entry.
Primary endpoint	Missegregation events per observed mitosis	Reproducible reduction versus vehicle within the Rescue Window.
Genomic safety	Micronuclei; karyotypic heterogeneity	No reproducible worsening.
Benchmark comparison	Magnitude/direction relative to WT-BUB1B rescue	Defines how much of genetically rescuable phenotype is pharmacologically approached.

Arm A falsification

Fingolimod fails the lead hypothesis if PP2A changes without kinetochore PP2A-B56 improvement and without reduced chromosome missegregation, or if apparent rescue is explained by toxicity, cytostasis, reduced mitotic entry, or mitotic arrest.

Why the genetic benchmark matters

If WT-BUB1B substantially rescues the patient-cell phenotype while fingolimod does not, the biological defect remains genetically rescuable but the pharmacologic bridge is inadequate. If both improve the same direct endpoint, the drug effect gains substantially stronger mechanistic interpretation.

4. Arm B discovery layer - from aneuploidy biology to patient-cell testing

Arm B asks a different question from Arm A: can a measured consequence of established aneuploidy be buffered without worsening chromosome instability? To broaden hypothesis generation without over-interpreting any single model, ARR5 adds a cross-species discovery layer using large-scale Saccharomyces cerevisiae aneuploidy datasets.

Why yeast belongs here

Aneuploid yeast provides a scalable system for identifying cellular programs associated with tolerance to chromosome-dosage imbalance. The framework focuses on buffering processes such as proteostasis, protein turnover, translation control, chaperone capacity, mitochondrial homeostasis and stress-response state. Muenzner et al. (Nature, 2024) profiled natural aneuploid isolates and reported extensive protein-level dosage compensation, supporting the use of yeast as a discovery system for dosage-buffering biology.

Cross-species rule

> Yeast aneuploidy signal → conserved human pathway? → phenotype present in BUB1B-MVA patient cells? → approved-drug mapping → patient-cell rescue + genomic safety

Yeast never constitutes therapeutic validation. A yeast association can only nominate a pathway. It advances only after human conservation is established and the corresponding abnormal phenotype is demonstrated in patient-derived cells. Drug efficacy is then judged in the human model, not inferred from yeast.

Discovery question	Translation gate	Human validation
Which processes track with aneuploid tolerance or dosage compensation?	Is the pathway conserved and mechanistically relevant to human aneuploid cells?	Is the corresponding phenotype abnormal in this child's cells?
Can the pathway be pharmacologically perturbed with an approved drug?	Does the drug engage the intended human target?	Does homeostasis improve without increased CIN, micronuclei or karyotypic heterogeneity?

Role in ARR5

This layer strengthens Arm B discovery while preserving the patient-specific decision tree. It does not alter the Arm A fingolimod hypothesis and does not create a third therapeutic arm. The sequence is PREDICT → PERTURB → VALIDATE → TRANSLATE, with patient-derived human cells as the decisive validation step.

5. Arm B - controlled buffering of established aneuploidy

MVA creates constitutional chromosome-dosage imbalance that may impose proteotoxic, metabolic and stress-response burdens. Arm B does not claim to correct BUB1B or reverse existing aneuploidy. It tests whether a demonstrated downstream cellular abnormality can be improved without permitting greater propagation of genomic instability.

Existing aneuploidy → measured stress phenotype → cross-species/human hypothesis generation → phenotype-matched approved drug → homeostasis improves → genomic safety preserved

4-PBA - lead Human fibroblasts with constitutional trisomies have shown proteostasis disturbance and protein aggregation, with 4-PBA reducing aggregate burden. Patient BUB1B-MVA cells must first demonstrate the relevant phenotype.	Arimoclomol - comparator An approved heat-shock/chaperone-response strategy provides a mechanistically distinct test of proteostasis buffering. It advances only if the relevant stress phenotype is present and genomic stability is preserved.

Candidate	Phenotype gate	Target engagement	Functional endpoint
4-PBA	Aggregation / folding stress	Aggregate burden; folding-stress markers	Improved homeostasis without increased CIN.
Arimoclomol	Proteostasis / heat-shock insufficiency	HSF1/HSP response	Improved homeostasis without increased CIN.
DRF / DMF	Oxidative-stress / Nrf2 phenotype	Nrf2 targets; redox state	Phenotype correction without increased CIN.
Elamipretide	Mitochondrial dysfunction	Mitochondrial function	Phenotype correction without increased CIN.

Genomic-safety rule

A drug that makes an aneuploid cell appear healthier while increasing missegregation, micronuclei, karyotypic heterogeneity, or expansion of unstable cells has failed.

6. Patient-specific experimental decision framework

The program follows the phenotype measured in this child's cells rather than a fixed treatment assumption. The same patient-derived model establishes the BUB1B rescue benchmark and determines which pharmacologic arm is biologically justified.

Observed patient-cell state	Experimental path	Required result
BUBR1-rescuable PP2A/CIN phenotype	WT-BUB1B benchmark → fingolimod Arm A	Target engagement plus reduced missegregation within Rescue Window.
Proteostasis phenotype	Yeast/human discovery support → 4-PBA + arimoclomol Arm B mini-screen	Correction of demonstrated homeostatic defect with preserved genomic stability.
Both phenotypes	Validate A and B independently	Only after single-agent success, test complementary combination.
Neither phenotype	Return to mechanism and drug landscape	No forced drug nomination.

Rescue Window

Improvement must occur without major loss of viability, extreme suppression of proliferation, reduced mitotic entry, artificial mitotic arrest, or worsening genomic instability. This prevents cytostasis from being misclassified as chromosome-fidelity rescue.

RESTORE - benchmark WT-BUB1B establishes whether the molecular phenotype is reversible and defines the genetic rescue ceiling. It is not an approved-drug candidate for this Track.	REBALANCE - Arm A Fingolimod tests downstream compensation of the BUBR1-PP2A-B56/kinetochore axis using a repurposed approved drug.	BUFFER - Arm B Yeast and human aneuploidy biology nominate buffering pathways; 4-PBA/arimoclomol test measured consequences in patient cells.

Future translational direction

Successful WT-BUB1B rescue would support future investigation of BUB1B restoration strategies, but systemic gene augmentation is outside this Track and remains limited by delivery, durability, tissue targeting and safety. The Track 2 candidates remain approved-drug repurposing hypotheses for preclinical testing.

7. Impact, innovation and scalability

This strategy separates three questions that are often conflated in rare-disease repurposing: whether the causal cellular phenotype is genetically reversible, whether an approved drug can compensate for the ongoing molecular defect, and whether the consequences of established aneuploidy can be safely buffered.

> Genotype → disease-relevant cellular phenotype → genetic rescue ceiling → cross-species/human discovery → approved-drug rescue → falsification & safety

Patient relevance

The decision tree terminates in this child's measured cellular biology. L737* expression, N1002K stability/function, BUBR1 abundance, kinetochore PP2A-B56, chromosome missegregation and stress phenotypes determine which candidate is tested and what constitutes rescue.

Innovation

The cross-species layer is deliberately asymmetric: yeast expands discovery power, while human patient-derived cells retain authority over translation. This allows large-scale aneuploidy biology to nominate conserved buffering pathways without treating yeast tolerance as evidence of human therapeutic efficacy.

Scalable unit

The transferable platform is genotype → molecular phenotype → genetic rescue benchmark → pathway discovery → approved-drug mapping → target engagement → direct functional rescue → genomic-safety confirmation. Drug identity remains disease- and phenotype-specific.

Translational boundary

These candidates are preclinical hypotheses, not treatment recommendations. No drug should be administered to the child on the basis of this analysis. Cellular rescue does not establish neurological, developmental, oncologic or systemic clinical benefit, and existing constitutional aneuploidy is not assumed to be reversible.

Conclusion

ARR5 proposes a focused, falsifiable repurposing program in which WT-BUB1B restoration defines the biological rescue benchmark, fingolimod tests pharmacologic rebalancing of ongoing chromosome-segregation error, and a cross-species discovery layer helps nominate controlled buffering strategies that must be validated in patient cells. A candidate advances only when molecular target engagement, direct functional rescue and genomic safety converge.

8. Evidence, reproducibility and disclosure

Selected evidence

1. Suijkerbuijk SJE et al. Molecular causes for BUBR1 dysfunction in Mosaic Variegated Aneuploidy. Cancer Res. 2010. PMID: 20516114.
2. Xu P et al. BUBR1 recruits PP2A via the B56 family of targeting subunits to promote chromosome congression. Biology Open. 2013;2:479-486. PMID: 23789096.
3. Baker DJ et al. Increased expression of BubR1 protects against aneuploidy and cancer and extends healthy lifespan. Nat Cell Biol. 2013;15:96-102. PMCID: PMC3707109.
4. Cordeiro MH et al. BUBR1 Pseudokinase Domain Promotes Kinetochore PP2A-B56 Recruitment, Spindle Checkpoint Silencing, and Chromosome Alignment. Cell Reports. 2020.
5. De Palma RM et al. FTY720/SET/PP2A experimental evidence supporting the pharmacologic bridge evaluated in Arm A. 2019. PMCID: PMC6529350.
6. Nawa N et al. Human constitutional aneuploidy/proteostasis work supporting the 4-PBA Arm B hypothesis. PLoS One. 2019. PMCID: PMC6663065.
7. North BJ et al. SIRT2 induces BubR1 and provides a mechanistic benchmark for BUBR1 stability biology. EMBO J. 2014. PMID: 24825348.
8. Muenzner J et al. Natural aneuploidy reveals proteomic dosage compensation across diverse yeast isolates. Nature. 2024. DOI: 10.1038/s41586-024-07442-9. Used for cross-species hypothesis generation, not direct human therapeutic evidence.

Cross-species evidence rule

Saccharomyces cerevisiae data are used to generate hypotheses about conserved aneuploidy-buffering processes. No yeast phenotype is treated as evidence of efficacy in BUB1B-MVA. Translation requires human pathway conservation, the corresponding phenotype in patient-derived cells, approved-drug target engagement, functional rescue and genomic-safety confirmation.

Reproducibility

The expanded approved-drug landscape, prioritization dimensions, experimental gates and cross-species discovery logic are maintained in the ARR5 repository and supporting screening files. Gated patient genomic data are not redistributed.

AI and data-handling disclosure

OpenAI ChatGPT Plus was used collaboratively for literature synthesis, code development, methodological review, adversarial hypothesis testing, scientific writing, and report development; the account-level 'Improve the model for everyone' setting was subsequently disabled.

Restore to benchmark. Rebalance the ongoing defect. Buffer only what is measured. Advance only when rescue and genomic safety agree.